

Unit 1

Introduction to Computer Networks.

NABIN THAPA

Course Content

1. Definitions, Uses, Benefits
2. Overview of Network Topologies (Star, Tree, Bus,...)
3. Overview of Network Types (PAN, LAN, CAN, MAN,...)
4. Networking Types (Client/Server, P2P)
5. Overview of Protocols and Standards
6. OSI Reference Model
7. TCP/IP Models and its comparison with OSI.
8. Connection and Connection-Oriented Network Services
9. Internet, ISPs, Backbone Network Overview

1.1. Definition ,Uses ,Benefits

- A computer network is a collection of computers or devices which are connected to each other for the purpose of sharing their resources.
- Any device which can share or receive the data is called a Node.
- Through which the information or data propagate is known as channels.
- It can be guided or unguided.



- ▶ A Collection of autonomous computers interconnected by a single technology.
- ▶ Two computers are said to be interconnected if they are able to exchange the information.
- ▶ Internet is the most known network of network.

Uses

- ▶ **File Sharing:** Networks offer a quick and easy way to share files directly.
- ▶ **Resource Sharing:** All computers in the network can share resources such as printers, fax machines, modems and scanners.
- ▶ **Communication:** Those on the network can communicate with each other via e-mail, messages etc.
- ▶ **Flexible Access:** Networks allow their users to access files from computers throughout the network.(It is an attempt to end the “**Tyranny** of geography”)
- ▶ **Sharing of Information:** Computer networks enable us to share data and information with the computers that are located geographically large distance apart.

Applications/Benefits

▶ Resource Sharing

- ▶ Hardware (computing resources, disks, printers)
- ▶ Software (application software)

▶ Information Sharing

- ▶ Easy accessibility from anywhere (files, databases)
- ▶ Search Capability (WWW)

▶ Communication

- ▶ Email
- ▶ Message Broadcast

Remote computing

Distributed processing

Application Area of Computer Network

Major applications areas of computer networks are:

1. Business Applications
2. Home Application
3. Mobile Computers
4. Healthcare and Medical Science
5. Education and Research
6. Entertainment
7. Research and Science
8. Environmental Monitoring etc.

Advantages and Disadvantages of Computer Network

Advantages:

- ▶ Offers Convenient resource sharing.
- ▶ Allows easy sharing of files.
- ▶ Provides the benefits of flexibility.
- ▶ Inexpensive to operate.
- ▶ Increase cost efficiency.
- ▶ Enhance communication and availability of info.

Disadvantages:

- ▶ Risk of Security issues
- ▶ Opens up a doorway for computer viruses and malware.
- ▶ Lacks Robustness.
- ▶ Requires efficient handler.
- ▶ Requires an expensive setup.

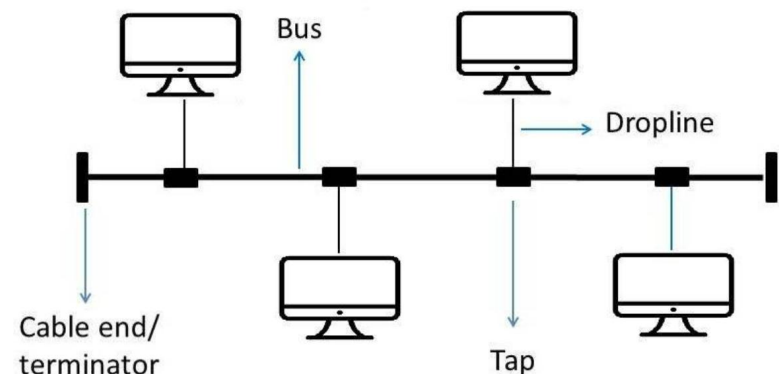
1.2. Overview of Network Topologies

Topology :Physical and logical Layout of network.(Arrangement of nodes)

- **Physical** – actual layout of the computer, cables and other network devices.(Geometric representation)/ placement of nodes.
- **Logical** – refers to the transmission of data between devices present in the network irrespective of the way devices are connected.
- **Some Common topologies:**
 - Bus
 - Star
 - Ring
 - Mesh
 - Hybrid
 - Tree
 - Wireless.

Bus Topology

- ▶ Uses single trunk or wires or backbone to which all of the computers on the networks are connected.
- ▶ Devices are connected to the backbone via drop lines and taps.
- ▶ Drop line: connects the device to the main cable.
- ▶ Tap: connector that splices into or punctures the main cable to create a connection.
- ▶ Data travels in both directions along the backbone. Only one device can transmit at a time to avoid collision.
- ▶ There is a limit on the number of taps and the distance between them due to signal degradation.



Bus Topology

Advantages

- ▶ Cheap and easy to implement
- ▶ Require less cable
- ▶ Does not used any specialized network equipment.
- ▶ One node failures can't affect the entire system.

Disadvantages

- ▶ Network disruption when computers are added or removed.
- ▶ A break in the cable causes system or whole system failures.
- ▶ Difficult to troubleshoot.

Star Topology

- ▶ Each device has a dedicated point-to-point link to a central controller called hub or switch.
- ▶ Each device requires a single cable
- ▶ Most widely implemented
- ▶ Devices are not directly linked to one another.
- ▶ Data traffic between devices goes through the hub.
- ▶ Each device needs only one link and one I/O port.
- ▶ Easy to install and reconfigure.
- ▶ Requires less cabling compared to mesh topology.
- ▶ Widely used in local-area networks (LANs).
- ▶ High-speed LANs often utilize a star topology with a central hub.

Star Topology

- Instead of hub device any of the following can be used:
layer 1 devices: hub or repeater, Layer2 device: Switch or bridge, layer 3 device: router or gateway.

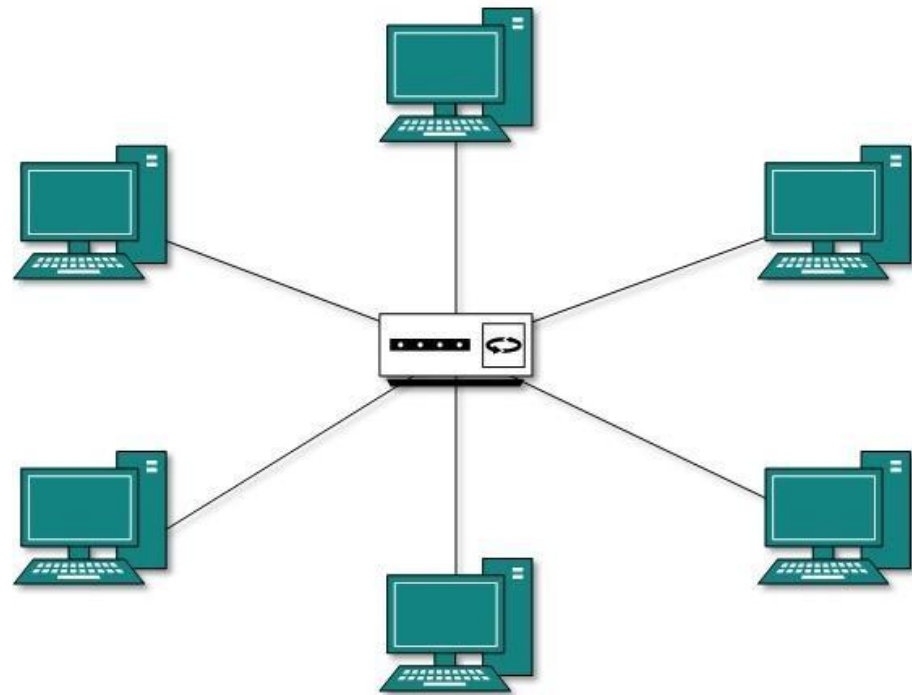


Fig: Star Topology

Star Topology

Advantages

- ▶ flexible/ Easy to setup and modification
- ▶ Robustness: Failure of one link does not affect others.
- ▶ Easy Fault Identification: The hub can monitor and isolate link problems.
- ▶ Simple Additions and Changes: Involves only connecting or disconnecting from the hub.

Disadvantages

- ▶ Expensive
- ▶ Performance is based on central device.
- ▶ Single Point of Failure: The entire system depends on the hub. If the hub fails, the whole network is down.
- ▶ Potentially More Cabling: Each node must connect to the hub, which can require more cabling than ring or bus topologies.

Ring Topology

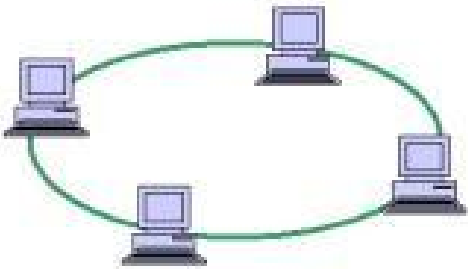
- ▶ Meaning that data travels in circular fashion from one computer to another on the network.
- ▶ Each device has a dedicated point-to-point connection with two adjacent devices.
- ▶ Signal passes in one direction, moving from device to device until it reaches its destination.(or two in dual-ring)
- ▶ Each device includes a repeater to regenerate and pass along the signal.
- ▶ Typically FDDI, SONET or Token Ring technology(only the device with the token can send data.) are used to implement a ring network.
- ▶ Ring networks are most commonly wired configuration.

1.FDDI (Fiber Distributed Data Interface):FDDI is a standard for transmitting data on **optical fiber** using **dual-ring** topology.

2. SONET (Synchronous Optical Network):SONET is a standard for **high-speed** digital data transmission over **optical fiber** in **telecommunication networks**.

Cont.

Ring Topology



Ring Topology

Advantages

- ▶ Cable failure are easily located, making troubleshooting easier.
- ▶ Ring Network are moderately easy of installation and Reconfiguration.
- ▶ No chance of data collision.

Disadvantages

- ▶ Expansion to the network can cause network disruption.
- ▶ A single break in the cable can disrupt entire network.

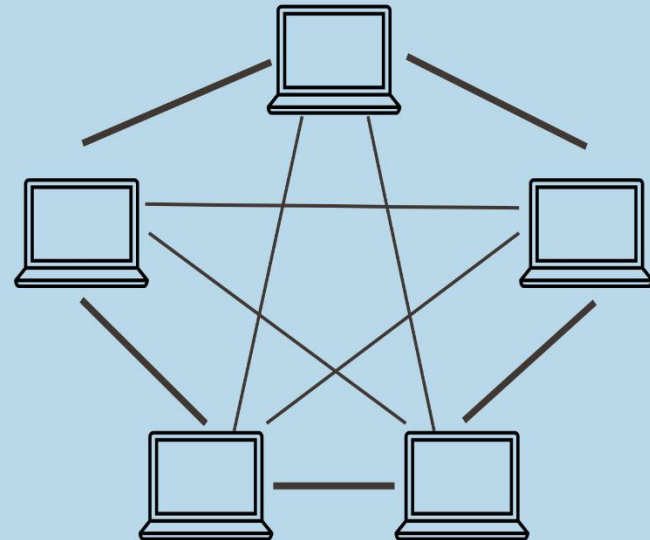
Mesh Topology

- ▶ Dedicated Links: Each device has a point-to-point link to every other device and Links carry traffic only between the two connected devices.
- ▶ Each computer connects to every other. High level of redundancy.
- ▶ Data can travel directly from sender to receiver without intermediary.▶ For n nodes, each node connects to $n-1$ others. Each device requires $n-1$ input/output ports.
- ▶ In duplex mode (bidirectional communication), total links: $n(n-1)/2$
- ▶ Rarely used.
 - Wiring is very complicated
 - Cabling cost is high
 - Troubleshooting a failed cable is tricky

Mesh Topology

Full Mesh: All hosts have a point-to-point connection to every other host in the network. Thus for every new host $n(n-1)/2$ connections are required. It provides the most reliable network structure among all network topologies.

Partially Mesh: Not all hosts have point-to-point connection to every other host. Hosts connect to each other in some arbitrary fashion. This topology exists where we need to provide reliability to some hosts out of all.



BOARD

Mesh Topology

Advantages

- ▶ The network can be expanded without disruption to current users device.
- ▶ Dedicated connections eliminate traffic problems.
- ▶ Robustness: one link failure doesn't affect the entire network.
- ▶ Security: dedicated lines prevent unauthorized access.
- ▶ Easy fault identification and isolation.

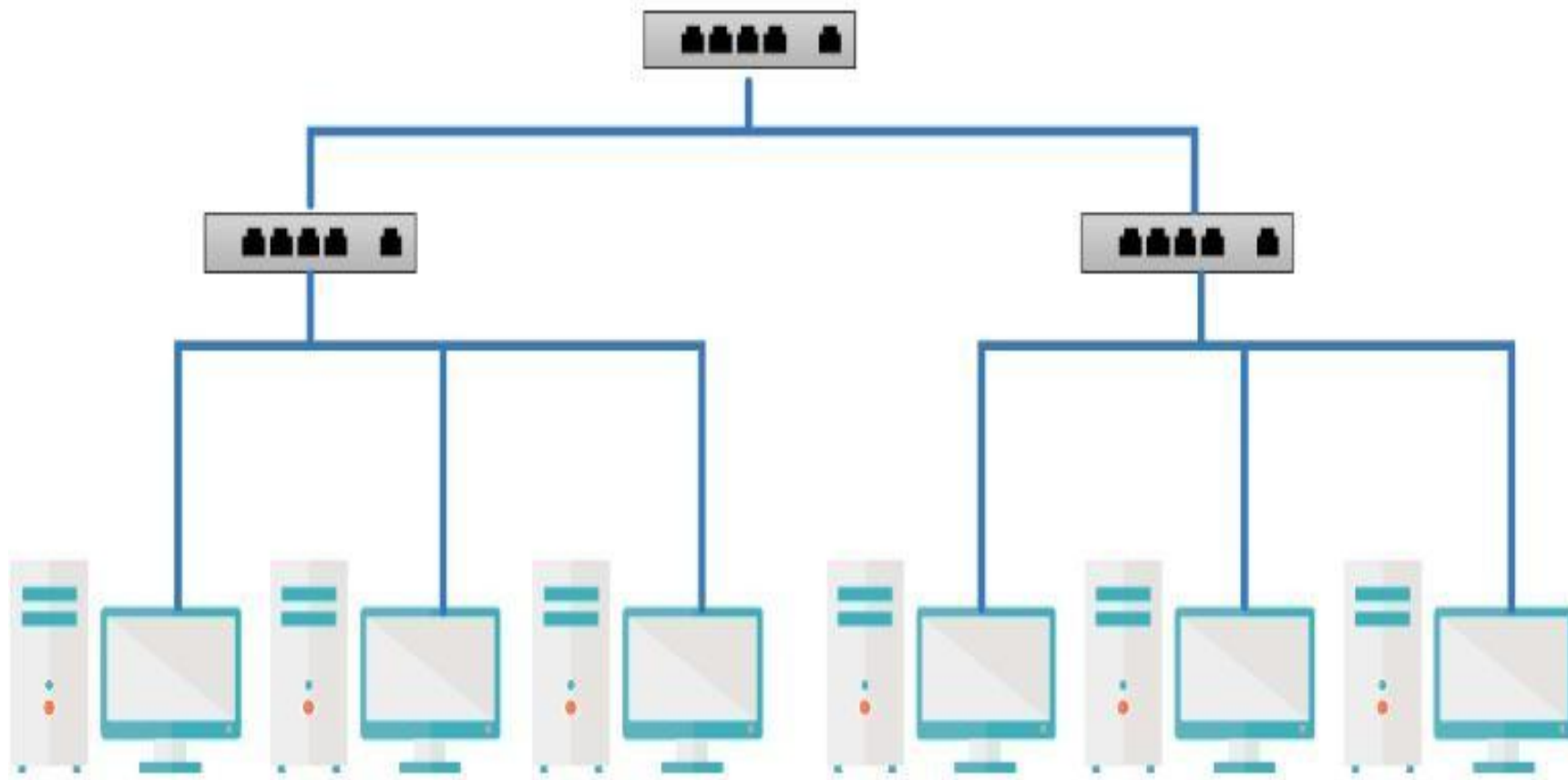
Disadvantages

- ▶ Require more LAN cable than other topologies.
- ▶ Complex installation and reconnection due to extensive cabling.
- ▶ Potential space issues for accommodating the bulk of wiring.
- ▶ High cost of hardware for I/O ports and cables.

Tree Topology

- ▶ In computer networking, tree topology is a type of network topology that resembles a tree(Also known as [Hierarchical Topology](#)).
- ▶ In a tree topology, there is one central node (the “trunk”), and each node is connected to the central node through a single path. Data flows from root to branches or vice versa.
- ▶ Tree topologies are often used to create large networks.

Tree Topology



Tree Topology

Advantages

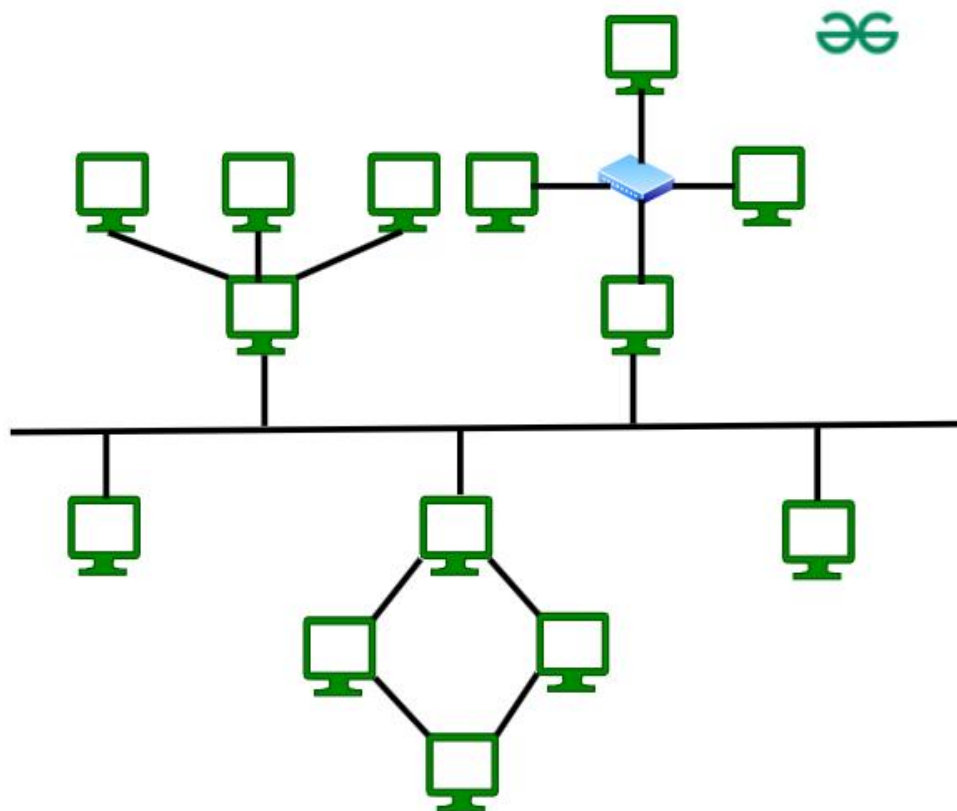
- ▶ It can support a large number of nodes.
- ▶ It can be easily expanded.
- ▶ Additionally, tree topologies are highly scalable and can be easily reconfigured.

Disadvantages

- ▶ It can be difficult to troubleshoot issues in a tree topology as each node is connected to multiple other nodes. This can make it hard to identify where the issue is located.
- ▶ Tree topology can be less reliable than other topologies such as star or mesh as there are more potential points of failure. If one node goes down, it can affect the entire network.
- ▶ Tree topology can be more expensive to implement as it requires more cabling and equipment than other topologies.

Hybrid Topology

- Hybrid topology is a type of network topology in which two or more different topologies are integrated or combined to lay out a network.



Advantages

- ▶ This topology is **very flexible** .
- ▶ The size of the network can be easily expanded by **adding new devices**.

Disadvantages

- ▶ The infrastructure cost is very high as a hybrid network **requires a lot of cabling and network devices** .
- ▶ **Hubs** used in this topology are **very expensive**.
- ▶ It is challenging **to design the architecture** of the Hybrid Network.

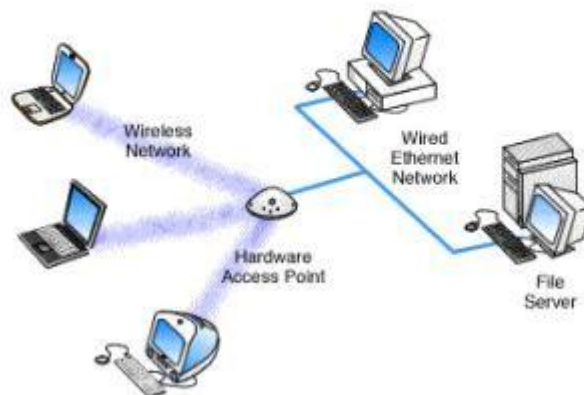
Compare Different types of network topologies

Wireless Topology

- ▶ Do not require physical cabling.
- ▶ Particularly useful for remote access for laptop users.
- ▶ Eliminate cable faults and cable breaks.
- ▶ Signal interference and security issue.

Wireless Topology

Wireless networking





Wireless Topology

Advantages

- ▶ Allows for wireless **remote access**.
- ▶ Network **can be expanded** without disruption to current users.
- ▶ No need for wired cable for local coverage.

Disadvantages

- ▶ Potential **security issues** associated with wireless transmission.
- ▶ **Limited speed** in comparison to other network topologies.
- ▶ External interference may arise.

1.3. Overview of Network Types

Types of Computer Networks (by their size / area / geographical area)

1. LAN(Local Area Network)
2. MAN(Metropolitan Area Network)
3. CAN(Campus Area Network)
4. PAN (Personal Area Network)
5. WAN(Wide Area Network)

PAN (Personal Area Network)

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- Personal area network (PAN) is an interconnection between different devices like smartphone, tablet, computer and other digital devices.
- PAN is used for a **personal purpose** like data sharing among devices and it has a **range of 10 meters**. Some type of personal area networks is **wired** like USB while others are wireless like Bluetooth.
- Wireless personal area network is also known as WPAN.

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LAN(Local Area Network)

- Network in small geographical Area (Room, Building or a School) is called LAN (Local Area Network)
- Connecting two or more personal computers .
- Less costly .
- Data is transferred at an extremely faster rate.
- Provides higher security.
- ▶ **Features of LAN:**
 - ▶ Cover small areas.
 - ▶ They are connected by Ethernet or Wi-Fi
 - ▶ Offer faster internet connection.

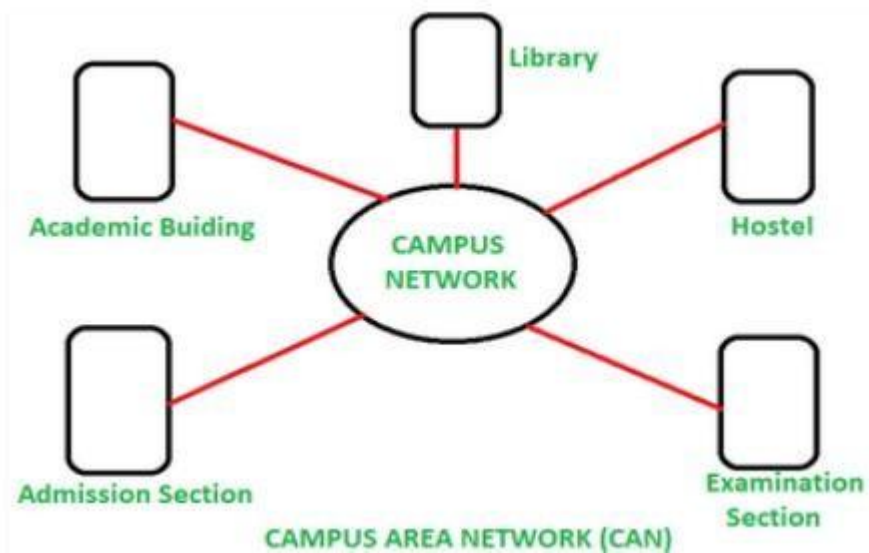


CAN(Campus Area Network)

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- Campus Area Network (CAN) is a group of interconnected Local Area Networks (LAN) within a limited geographical area like school campus, university campus, military bases, or organizational campuses and corporate buildings etc.
- This Campus Area Network also called as Corporate Area Network.
- This network is also referred as Residential Network or ResNet as it is only used by residents of specific campus only.

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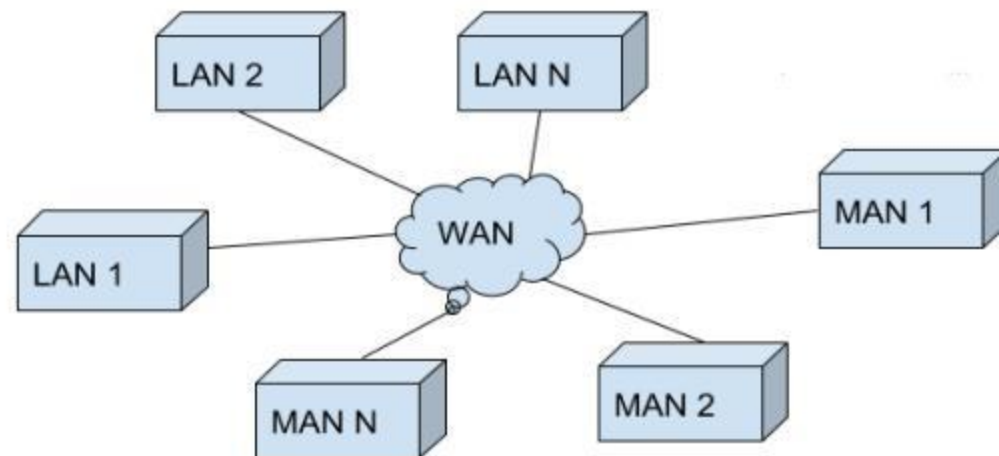
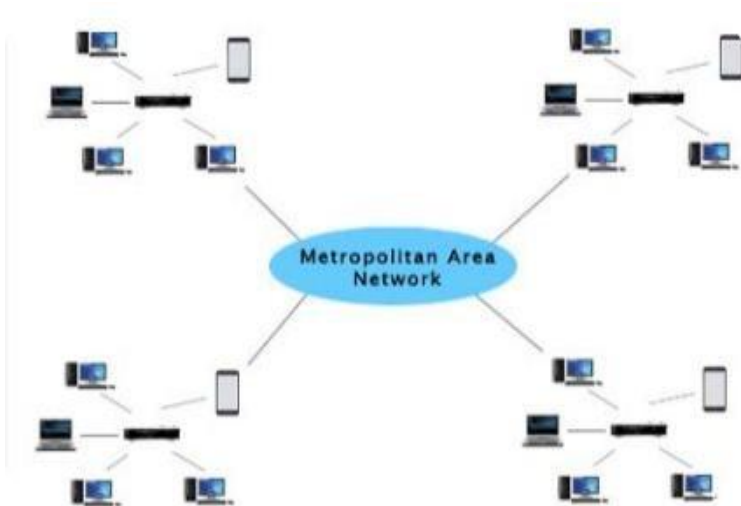


MAN(Metropolitan Area Network)

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- Network in a City is called MAN (Metropolitan Area Network)
- MAN network **covers larger area by connections LANs** to a larger network of computers.
- The size of the MAN is **larger than LANs and smaller than WANs** (wide area networks), a MANs covers the larger area of a city or town.
- ▶ **Features of MAN:**
 - ▶ Span cities, buildings, and towns.
 - ▶ Connected by modems and cables.
 - ▶ Offer average internet connection.

Contd.



WAN(Wide Area Network)

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- A Wide Area Network is a network that **extends over a large geographical area such as states or countries.**
- A WAN is quite bigger network than the MAN.
- A WAN is not limited to a single location, but it **spans over a large geographical area through a telephone line, fiber optic cable or satellite links.**
- **The internet is one of the biggest WAN in the world.**
- A WAN is widely used in the field of Business, government, and education.

Question: Explain PAN, LAN, CAN, MAN, WAN with advantages and disadvantages.

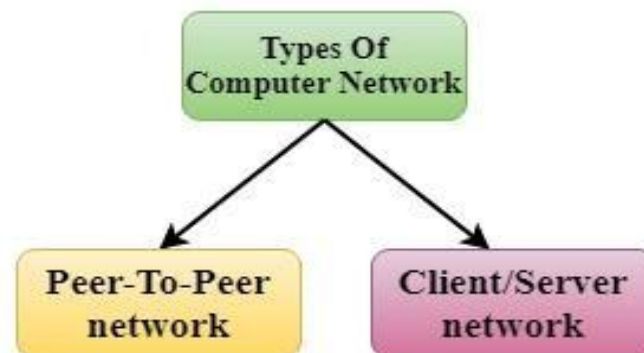
Storage Area Network (SAN)

- A Storage Area Network (SAN) is a fast and specialized network that connects servers to storage devices. It helps large organizations store and access data quickly and efficiently.
- Example: Bank Data Center
- A big bank uses a SAN in its data center to handle millions of daily transactions. To keep customer data safe and easy to access, the bank uses a Fibre Channel SAN. This setup allows their servers to quickly connect to storage systems and retrieve information when needed.

Parameter	PAN	LAN	CAN	MAN	WAN
Full Form	Personal Area Network	Local Area Network	Campus Area Network	Metropolitan Area Network	Wide Area Network
Range	1-100 meter (Max)	Upto 2KM	1-5KM	5-50KM	Above 50KM
Transmission Speed	Very High	Very High	High	Average	Low
Area	Within in a room	Within in office ,buildings	Within university ,corporate offices	Within City	Within in countries.
Ownership	Private	Private	Private	Private or Public	Private or Public
Maintenance	Very Easy	Easy	Moderate	Difficult	Very Difficult
Error Rate and Cost	Very low	Low	Moderate	High	Very High

1.4. Networking Types(Architecture)

- Computer Network Architecture is defined as the **physical and logical design of the software, hardware, protocols, and media of the transmission of data.**
- Architecture also defines **how the computers should get connected to get the maximum advantages of a computer network such as better response time, security, scalability** et C.
- Generally two types:
 - ▶ Peer-To-Peer network
 - ▶ Client/Server network



Peer-Peer Architecture

- “Peers” are computer systems which are connected to each other.
- Files can be shares directly between systems on the network **without the need of central server.**
- focus on the connectivity.
- Each computer on **P2P** network **becomes a file server as well as a client.**
- Once connected to N/w,P2P software allows you to search for files on other clients computers.

Peer-Peer Architecture

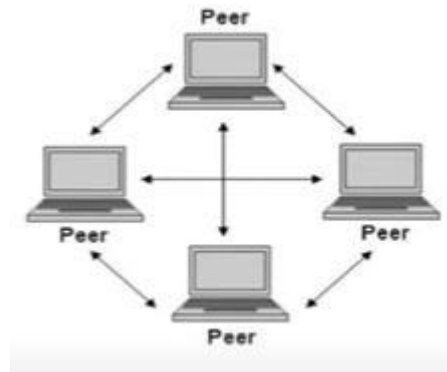


Fig: Peer-Peer Architecture

Advantages of a Peer to Peer Architecture

- **Less costly** : **no central server** that has to take the backup.
- **Robust**: In case of a computer failure all other computers in the **network are not affected** and they will continue to work as same as before the failure.
- **Installation of peer to peer architecture is quite easy** **as each computer manages itself.**
- The server is not **bottlenecked** as the services are provided by the several nodes.

Disadvantages of a Peer to Peer Architecture

- Each computer has to take the **backup** rather than a central computer **and** the **security measures are to be taken by all the computers separately.**
- **Scalability is an issue in a Peer to Peer** Architecture as connecting each computer to every computer is a headache on.

Client/Server network(Tired)

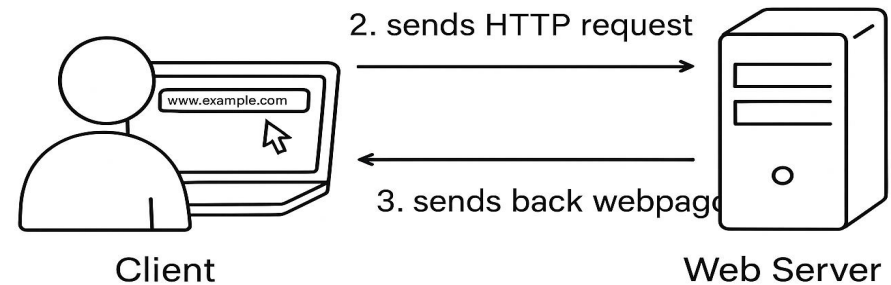
- ▶ A network architecture in which each computer or node on the network is either a client or server.
- ▶ **Server:**
 - ▶ Provides services for clients.
 - ▶ Also known as a **host computer**.
 - ▶ Controls access h/w and s/w on the network.
 - ▶ Provide storage area for programs, data and information.
 - ▶ Provide **access** to stored data.
- ▶ Server may be Application Server, Message Server, Proxy Server, Database Server, Web Server etc.
- ▶ System administration manages the data on the server.

Client/Server network

Client:

- ▶ Also known as **user or computer or end user**.
- ▶ **Request** the resources from the server.
- ▶ **Independent** from other clients
- ▶ **Gets the file from server**
- ▶ **Sends file to the server**
- ▶

1. A user types a website URL in a browser



Example Workflow (Web Browser)

Advantages of Client Server Architecture

- **Data backup is easy and cost effective** as there is no need to manage the backup on each computer.
- **Performance is better** as the response time is greatly improves because the server is more powerful computer than the other computers in the network.
- **Security is better as unauthorized access are denied** by server computer and all the data goes through the server.
- **Scalability is not an issue in this Architecture** as large number of computers can be connected with server.
- Use of **dedicated server increases the speed** of sharing resources.
- Centralization control.

Disadvantages of Client Server Architecture

- ▶ Requires **specialized server** with large memory and s and secondary storage. leads to the **increase in cost**.
- ▶ Require special operating system: **Network Operating system**
- ▶ Requires **Dedicated Network Administrator**
- ▶ There may be a chance of server getting **bottlenecked**.
- ▶ **Expensive**
- ▶ **If the server fails, whole n/w goes down.**

Aspect	Peer-to-Peer (P2P)	Client-Server
Network Structure	All computers are equal peers in the network.	Computers have different roles: some are clients, and some are servers.
Node Roles	All computers can act as both clients and servers.	Computers have distinct roles - clients make requests, servers fulfill them.
Resource Sharing	Resources are shared directly among peers.	Resources are centralized on servers, and clients request them.
Scalability	Easily scalable; new peers can join without central coordination.(Performance can drop)	Scalability can be complex due to server capacity constraints.(But load balance by server)
Control and Security	Less control and security, as it's challenging to enforce access restrictions in a decentralized network.	More control and security, with centralized access control on servers.
Expense	Less expensive to implement	Expensive
Reliability	Can be less reliable; if many peers go offline, network performance suffers.	More reliable; redundancy and backups can be managed centrally.
Example Applications	File sharing (e.g., BitTorrent), blockchain, .	Web services, email servers, database servers, enterprise applications.
Ease of Setup	Easier setup; minimal infrastructure	More complex setup; requires server configuration and maintenance

1.5 . Overview of Protocols and Standards

- Two entities can not simply send bit streams to each other and expect to be understood. For communication to occur, the entities **must agree on a protocol**.
- A protocol is a **set of rules that governs(control) data communications**.
- A protocol defines **what is communicated, how it is communicated, and when it is communicated**.
- The key elements of a protocol are :
 - ▶ Syntax
 - ▶ Semantics
 - ▶ Timing

Elements of Protocols

► Syntax:

- **Structure or format** of the data, meaning the **order** in which they are presented
- Indicates how to read the bits - field border or boundary.
- **For example, a simple protocol might expect the first 8 bits of data to be the address of the sender, the second 8 bits to be the address of the receiver, and the rest of the stream to be the message itself**
- **Syntax should be same in sender and receiver for to communicate.**

► Semantics:

- Interprets the **meaning** of each section of bits.
- How is a particular pattern to be interpreted, and what action is to be taken based on that interpretation.
- **For example, does an address identify the route to be taken or the final destination of the message?**

► **Timing:**

- **When data should be sent ?**
- Speed at which data should be sent or speed at which it is being received.
- **For example, if a sender produces data at 100 Mbps but the receiver can process data at only 1 Mbps, the transmission will overload the receiver and some data will be lost.**

Protocols examples:

- ▶ **Transmission Control Protocol (TCP):** TCP is a transport layer protocol which provides end-to-end reliable communication , error control and flow control services..
- ▶ **Internet Protocol (IP):** IP is network layer protocol. It is mostly used with TCP. The IP addresses in packets help in routing them through different nodes in a network.
- ▶ **User Datagram Protocol (UDP):** UDP is a substitute communication protocol to (TCP) Transmission Control Protocol..
- ▶ **Post office Protocol (POP):** It is an application layer protocol. POP3 is designed for receiving incoming E-mails.
- ▶ **Simple mail transport Protocol (SMTP):** It is an application layer protocol.SMTP is designed to send and distribute outgoing E-Mail.
- ▶ **File Transfer Protocol (FTP):**It is one of the oldest application layer protocol. FTP allows users to transfer files(program files, multimedia files, text files, and documents, etc.) from one machine to another.
- ▶ **HyperText Transfer Protocol (HTTP):** It is most commonly used application layer protocol for transmitting hypertext document in internet. It provides a platform for data communication in www.
- ▶ **HyperText Transfer Protocol Secure (HTTPS):** It is a standard protocol to secure the communication among two computers in www. It is the combination of HTTP and some security standards..
- ▶ **Telnet:** It is remote login protocol for executing command on a remote host. it is based on client server mode.A client provides Telnet request to the telnet server.(application layer protocol)

Standards

- Standards **provide guidelines to manufacturers, vendors, government agencies, and other service providers** to ensure the kind of interconnectivity necessary in today's marketplace and in international communications.
- Standards are essential in creating and maintaining an open and competitive market for equipment manufacturers and in guaranteeing.
- Data communication standards fall into two categories:
- De facto and De jure
- **De facto.:**

Standards that **have not been approved** by an organized body **but have been adopted as standards** through widespread use are de facto standards. **De facto standards are often established originally by manufacturers who seek to define the functionality of a new product or technology.**

De jure.

Those standards that **have been legislated by an officially recognized body** are de jure standards.

Standard , Forums and Open standard

Some standards are:

- International Organization for Standardization (ISO)
- International Telecommunication Union-Telecommunication Standards Sector (ITU-T)
- American National Standards Institute (ANSI)
- Institute of Electrical and Electronics Engineers (IEEE)
- Electronic Industries Association (EIA)

Forums:

- Industry groups that develop and promote technologies collaboratively before they become official standards.
- Example: Wi-Fi Alliance, Bluetooth SIG.

Open Standards:

- Publicly available and free-to-use specifications developed through open consensus.
- Example: TCP/IP, HTML, PDF, OSI.

1.6 . OSI Reference Model

- OSI- Opens System Interconnections.
- It was proposed by ISO (International Organization for Standards) in 1984 after the TCP/IP model.
- Starts developing in late 1970s and Approved by 1984.
- The term “**Open**” in Open System Interconnections denotes “**to communicate with any 2 systems**”.
- There are 7 layers in OSI Reference model.
- It is also called OSI layered architecture / OSI Protocol architecture.
- **The process of breaking up the functions or tasks of networking into layers reduces complexity.**
- Each layer **provides a service to the layer above it** in the protocol specification.
- **Each layer communicates with the same layer’s software or hardware on other computers.**

Layers of OSI Reference Model

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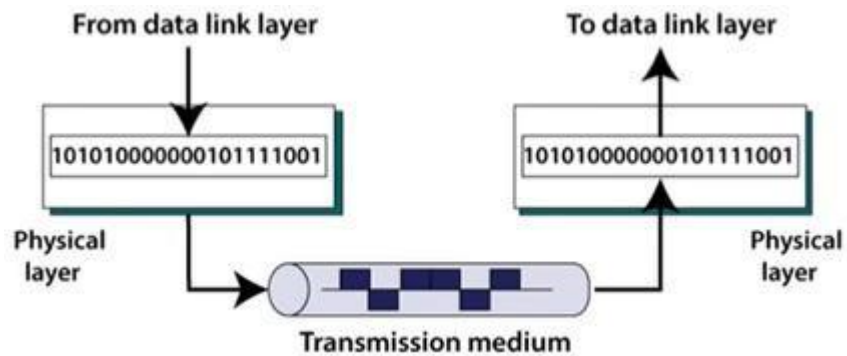
OSI Model

- | | | | |
|----|---------------------|-------------|---|
| 07 | Application | • - - - - - | The closest layer to the user; provides application services. |
| 06 | Presentation | • - - - - - | Encrypts, encodes and compresses usable data. |
| 05 | Session | • - - - - - | Establishes, manages, and terminates sessions between end nodes. |
| 04 | Transport | • - - - - - | Transmits data using transmission protocols including TCP & UDP. |
| 03 | Network | • - - - - - | Assigns global addresses to interfaces and determines the best routes through different networks. |
| 02 | Data link | • - - - - - | Assigns local addresses to interfaces, delivers information locally, MAC method |
| 01 | Physical | • - - - - - | Encodes signals, cabling and connectors, physical specifications. |

Physical Layer

- In the OSI model, the physical layer is the **lowest layer**, and it **deals with the actual physical transmission** of data **over a physical medium**, such as copper wires, optical fibres, or wireless radio waves.
- It **defines the hardware characteristics** and **electrical/optical specifications** of the transmission medium.
- Examples of devices at the OSI physical layer include **Hub, Repeater, Modem, Cables**.

Physical Layer



Data Link Layer

- ▶ It receives the data from network layer and creates frames, and add physical address to these frames.
- ▶ The data unit in the data link layer is called Ethernet frame.
- ▶ The DLL is divided into two sub layers.
 - ▶ MAC(Media Access Control).
 - ▶ LLC(Logical Link Control).
- ▶ The MAC sub-layer is responsible for data encapsulation and accessing media.
- ▶ **Encapsulation:** In encapsulation IP packet receives from network layers are added with header and trailer.

Data Link Layer

► Header:

- Contains MAC address of sender and Receiver.

► Trailer:

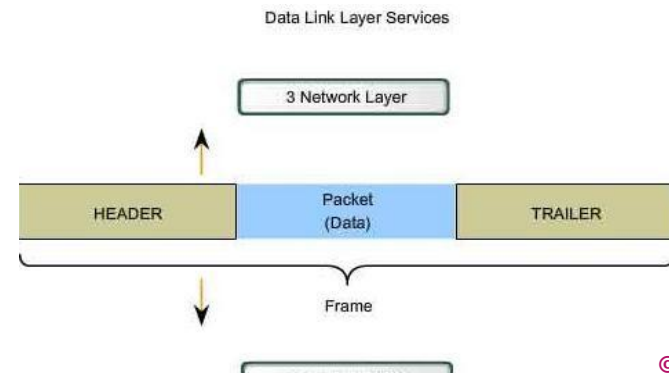
- 4 bytes of error checking data used to detect error in received Ethernet frame.

► Access Method :

CSMA/CD(Carrier-sense multiple access with collision detection)

► LLC(Logical Link Control).

- It offers flow control and error control.



DLL is responsible for :

- ▶ **Framing**: Divides data into manageable units called frames.
- ▶ **Physical Addressing**: Adds sender and receiver MAC addresses to each frame.
- ▶ **Flow control**: Manages data rate to prevent overwhelming the receiver.
- ▶ **Error control**: Detecting and Retransmitting damaged or lost frame
- ▶ **Access control**: Which device has control over the link ?

Network Layer

- The transport layer passes the TCP segments or UDP datagram to the network layers.
- Then NL accept the transport layer segments and **add logical addressing** and form a **packet**.
- Responsible **for moving packets from source to destination** accross the multiple link.
- Also **determine best path** for packet delivery.
- It offers
 - **Logical addressing**:Ip addressing
 - **Routing**:Responsible for delivering packets to the final destination
 - **Path determination** : choosing the best path(eg: OSPF: open shortest path first)

Transport Layer

- Transport layer ensure **process to process delivery of message**.
- **It uses TCP and UDP protocols.**
- When the message reaches the transport layers, one transport layer protocols TCP or UDP is selected.
- Normally, OSI transport layer is connection oriented . i.e. TCP is selected.

TCP

- Supports segmentation
- Connection oriented
- Reliable
- TCP segment

UDP

- Does not support segmentation
- Connection less
- Unreliable
- UDP datagram

Transport layer Responsible for :

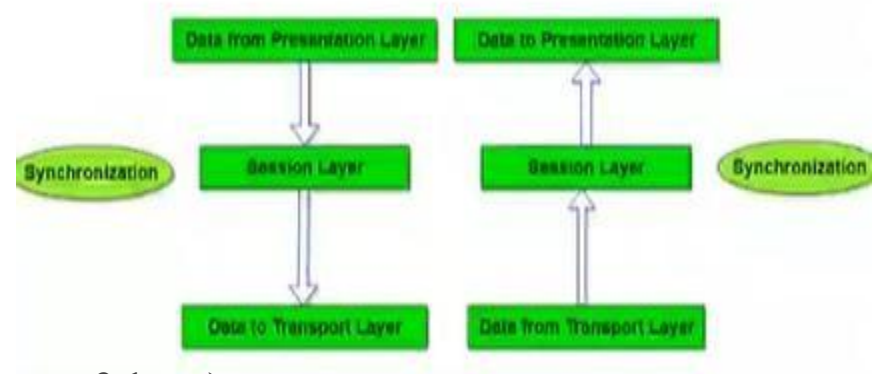
- ▶ **Port Addressing**: specifies a specific process or service eg: HTTP: 80,FTP: 21, SMTP 25
- ▶ **Segmentation and reassembly**: message are divided into segment
- ▶ **flow and error control**:
- ▶ **Connection control**: This can either be connection oriented or connectionless

Session Layer

- Creates communication channel and controlling ports and session.
- It establishes, maintains, and ends a session.
- It is also responsible for **dialog control** (either in half duplex or full duplex) and Co-ordinates communication between the system.
- Session layer enables two systems to enter into a dialog/Communication.

- Session management.
- Authentication.
- Authorization.
- Synchronization

(Allow to add checkpoints in to a stream of data)



Presentation Layer

- Presentation Layer is **responsible for to Translate, Compress, Encrypt and Decrypt data.**
- The data is in the form of character or numbers.
- Ex: Character code **translation** from ASCII to EBCDIC
 - ▶ ASCII → EBCDIC (Extended Binary Coded Decimal Interchange Code)
- **Before data is transmitted, presentation layer reduces the number of bits that are use for represent original data. This bit reduction process is called compression.**
- It helps to **encrypt** data for security purpose i.e. it enhance security.

Application Layer

- Application layer **allows access to network resources.**
- In the application layer, an application, such as an internet browser, gets the data and a user can then interact with it.
- Application-layer helps to identify communication partners, determining resource availability, and synchronizing communication.
- It allows users to log on to a remote host.
- This layer provides various e-mail services.
- Highest level of OSI layers also called closest layer from user.

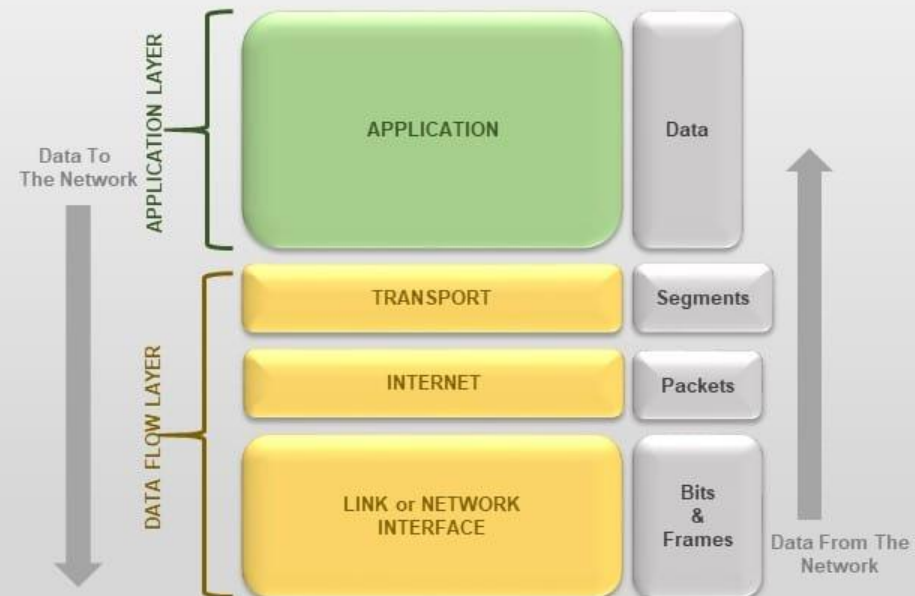
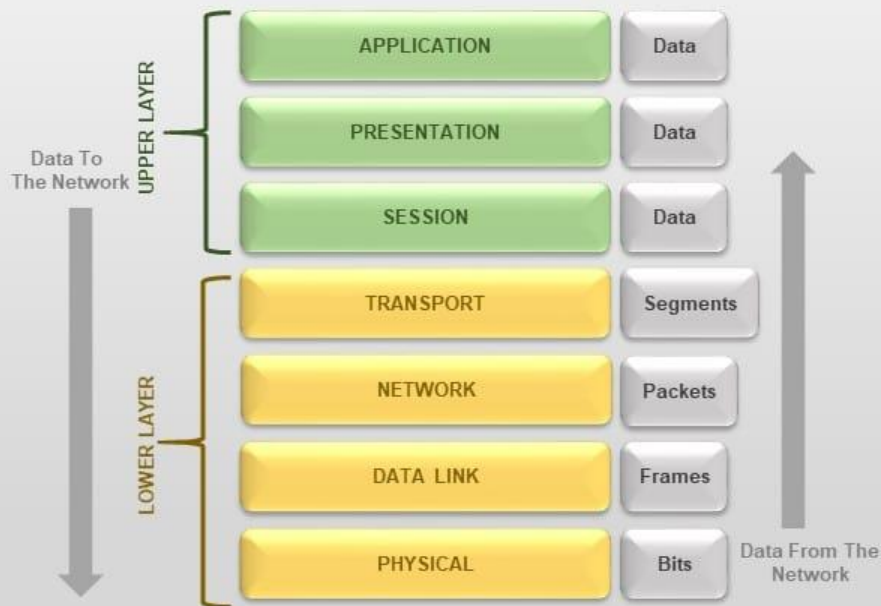
Protocols used in OSI reference Model

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Layer	Name	Protocols
Layer 7	Application	SMTP, HTTP, FTP, POP3, SNMP, DNS etc.
Layer 6	Presentation	MPEG, ASCII, SSL, TLS.
Layer 5	Session	NetBIOS, PPTP (Point-to-Point Tunneling Protocol)
Layer 4	Transport	TCP, UDP
Layer 3	Network	IPV4, IPV6, Internet Control Message Protocol (ICMP) , IPSEC, ARP.
Layer 2	Data Link	PPP, Frame Relay, ATM, etc.
Layer 1	Physical	RS232, 100BaseTX, ISDN.

Mapping TCP/IP with OSI model

OSI MODEL vs TCP/IP MODEL



1.7.TCP/IP Model and its Comparison with OSI

Brief History of TCP/IP Model

- ▶ TCP/IP protocol suite was **developed before the OSI model**.
- ▶ TCP/IP is a set of protocols developed **to allow co-operating computers** to share resources across a network.
- ▶ In 1969 the Defense Advanced research projects Agency (DARPA) funded a research and development project to **create an experimental packet switching network. This network is called ARPANET**.
- ▶ In 1975 the ARPANET was converted from an experimental network to an operational network, and the responsibility for administering the network was given to the Defense Communication Agency (DCA).

Cont.

- ▶ The TCP/IP protocols **were adopted as Military Standards** (MIL STD) in 1983, and all hosts connected to the network were required to convert to the new protocols.
- ▶ DARPA funded to implement TCP/IP in Berkely Unix.
- ▶ In 1983, the old ARPANET was divided into MILNET and smaller ARPANET. The Internet was used to refer to the entire network; MILNET and ARPANET.

TCP/IP Model

- The TCP/IP model is a widely used networking model that consists of four layers: Application, Transport, Internet, and Link(Network Access Layer).
- TCP/IP is a set of protocols developed to allow cooperating computers to share resources across a network.
- It serves as the foundation for the modern internet and is responsible for the end-to-end communication of data packets across networks.
- The TCP/IP model combines the OSI model's physical and data link layers into a single "Link Layer" due to its focus on real-world networking.
- Key protocols within the TCP/IP model include TCP (Transmission Control Protocol) and IP (Internet Protocol), which enable reliable communication and routing of data packets.
- It is a more flexible and adaptable model compared to OSI, making it well-suited for the dynamic and evolving nature of the internet.

TCP/IP Model

The TCP/IP Model



TCP/IP Model

Layers

- ▶ Application Layer
- ▶ Transport Layer
- ▶ Internet Layer
- ▶ Network Access Layer/Link layer

PDUs(Protocol data unit)

- ▶ Application message
- ▶ TCP segment/UDP datagram
- ▶ IP packet
- ▶ Ethernet Frame

TCP/IP Model

▶ Application Layer:

- ▶ This layer is responsible for **providing network services directly to end-users or applications**. It includes protocols like **HTTP** (for web browsing), **SMTP** (for email), **FTP** (for file transfer), and more. Application layer protocols **facilitate communication between software applications running on different devices**.

▶ Transport Layer:

- ▶ The transport layer is **responsible for end-to-end communication and data flow control** between devices on different networks. **It includes two main protocols:** Transmission Control Protocol (TCP) and User Datagram Protocol (UDP). TCP provides reliable, connection-oriented communication, while UDP offers connectionless, unreliable communication.

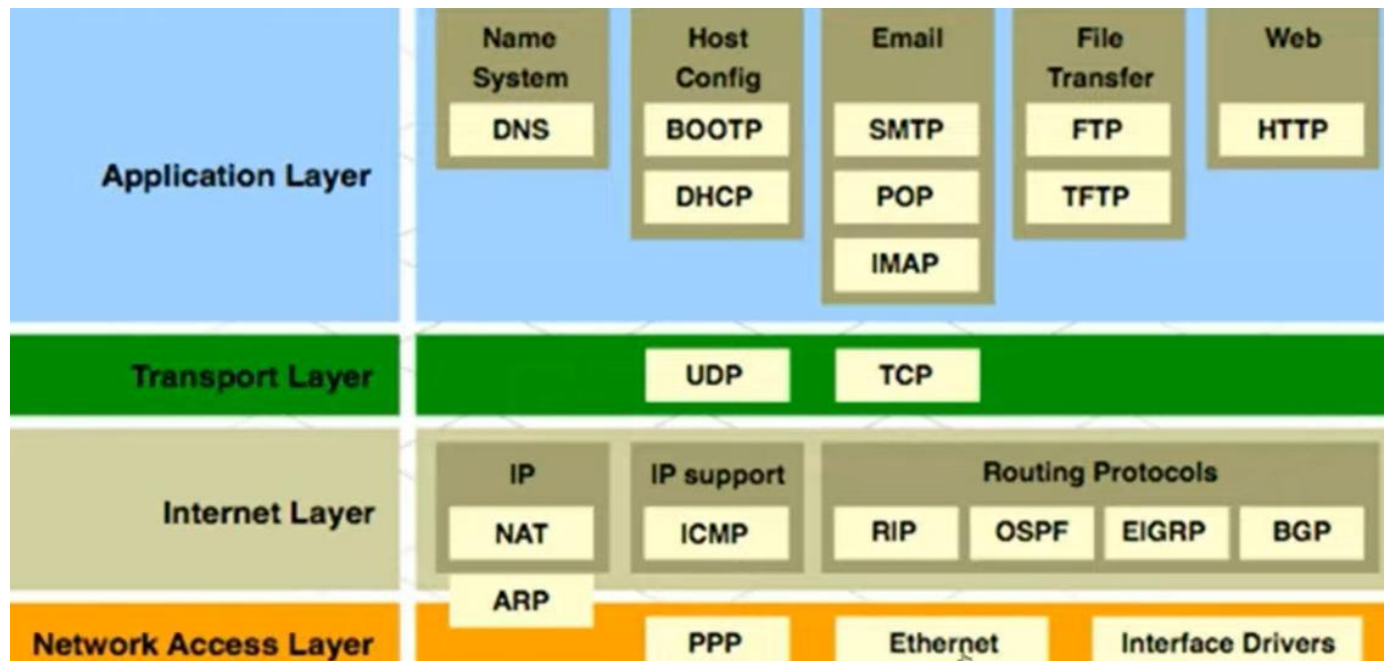
Internet Layer:

- ▶ This layer **handles logical addressing, routing, and forwarding of data packets** across different networks. The **primary protocol at this layer is the Internet Protocol (IP)**, which enables the routing of packets from the source to the destination based on IP addresses.

Link Layer:/Network Access Layer

- ▶ The link layer, **sometimes called the network interface or data link layer**, deals with the physical connection between devices **on the same local network segment**. It includes hardware-specific protocols and technologies, such as Ethernet, Wi-Fi, and others. This layer is responsible for **addressing and error detection at the local network level**.

Protocols Used in TCP/IP Model



Parameters	OSI Model	TCP/IP Model
Full Form	OSI stands for Open Systems Interconnection.	TCP/IP stands for Transmission Control Protocol/Internet Protocol.
Layers	It has 7 layers.	It has 4 layers.
Usage	It is low in usage. Generally in Research.	It is mostly used for network design.
Model	It is theoretical/Reference Model.	It is Practical/Implementation Model.
Delivery	Delivery of the package is guaranteed in OSI Model.	Delivery of the package is not guaranteed in TCP/IP Model.
Horizontal/ vertical Approach	Follows vertical approach	Follows Horizontal approach : the Transport layer on one device communicates directly with the Transport layer on another device to ensure end-to-end data delivery.
Reliability	It is less reliable than TCP/IP Model.	It is more reliable than OSI Model.
Developed By	ISO(International Organization for Standardization)	DoD(Department of Defense)
Replacement	Protocols are hidden in OSI model are easily replaced as the technology changes	In TCP/IP replacing protocols is not easy
Layer Separation	OSI model has separate Presentation layer and Session layer.	TCP/IP model does not have separate Presentation layer and Session layer.

Assignment

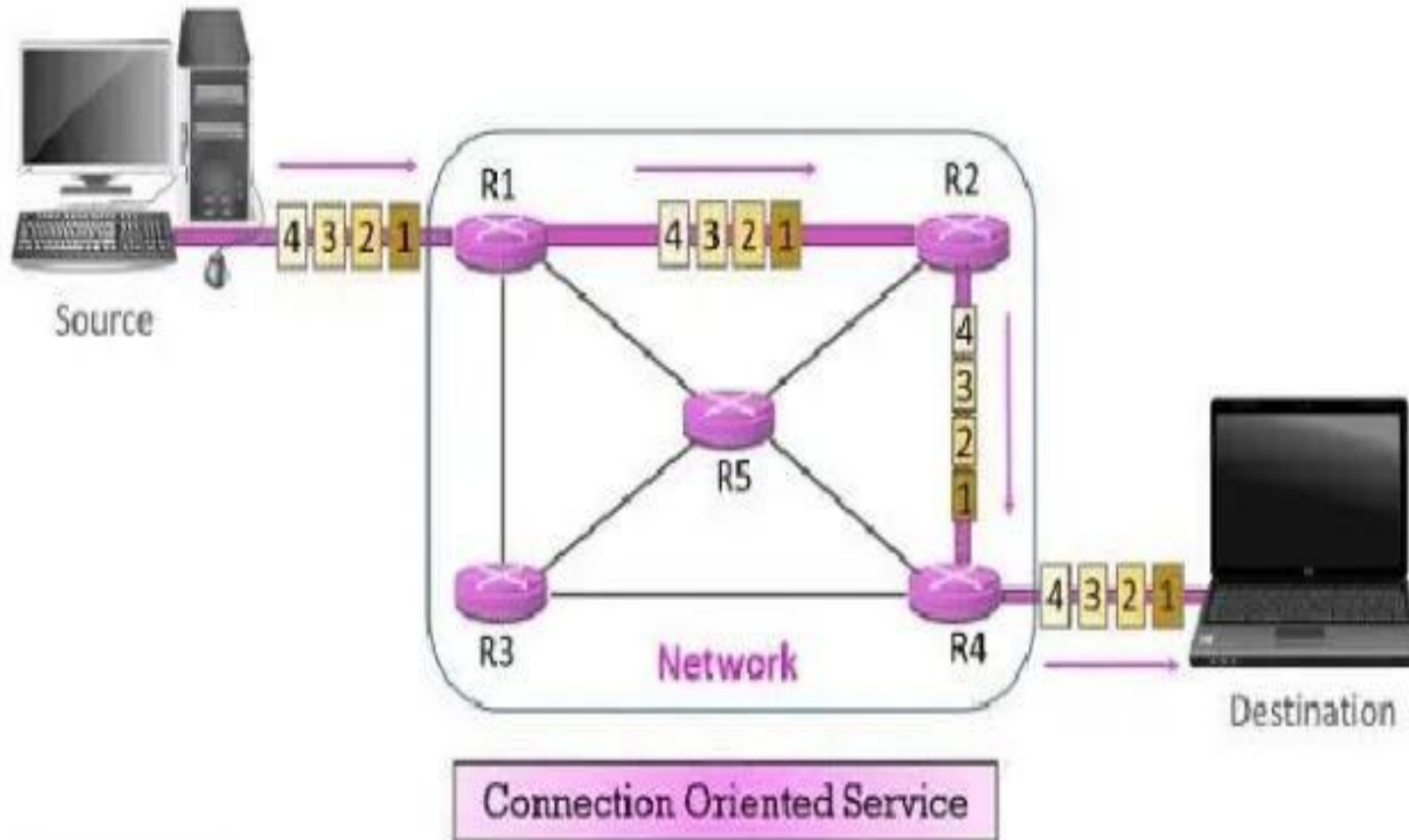
1. Explain the Encapsulation and Decapsulation process in Internet model.
2. Explain relationship of layers and addressing in TCP/IP.

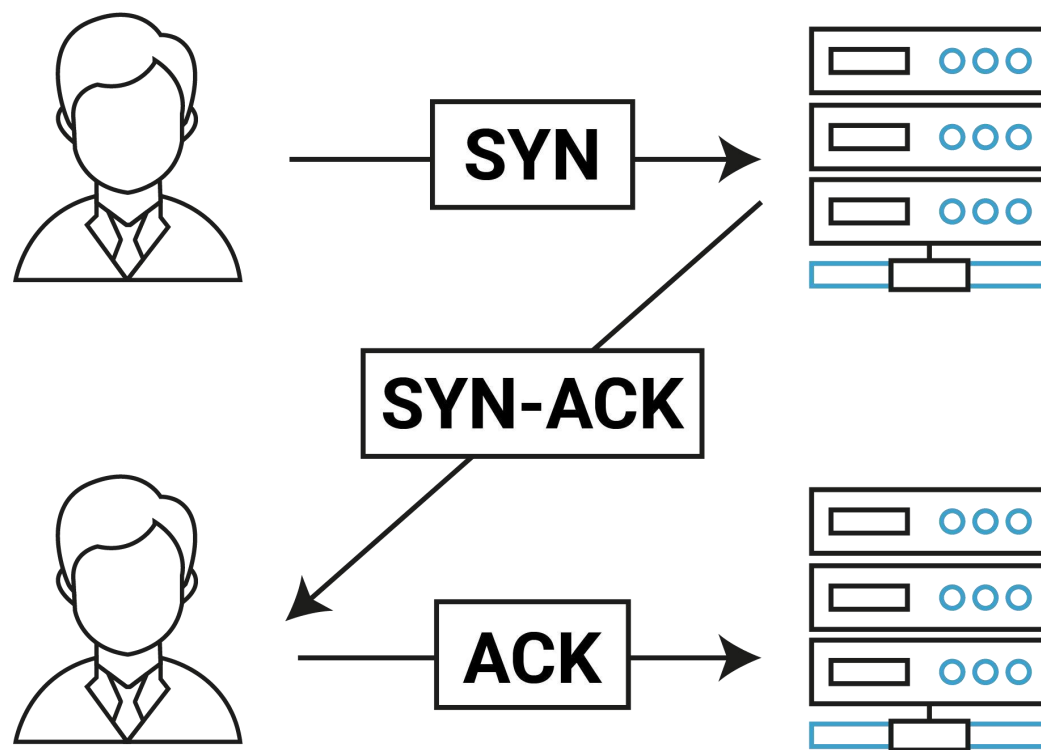
1.8.Connection and Connection –oriented Network Service

A "**connection**" often refers to **a link or bond** between two or more things/ Devices.

- For example, **electrical connections, internet connections, or the links between different parts of a machine.**
- Before communication ,the user in connection oriented service follows following sequence of operations.
 - **Connection is established.(Virtual Connection)**
 - **Data/Information Sent.**
 - **Connection Release.**

Connection Oriented-Service





TCP 3-Way Handshake (Steps):

- **SYN (Client → Server)**
Client sends SYN to request connection.
- **SYN-ACK (Server → Client)**
Server replies with SYN-ACK to accept and acknowledge.
- **ACK (Client → Server)**
Client sends ACK to confirm, and the connection is established.

Purpose:

- Establish a reliable connection.
- Synchronize sequence numbers.
- Ensure both sides are ready to communicate.

Cont.

- ▶ Connection Oriented service is more reliable than connectionless services.
- ▶ Common example of Connection oriented service is TCP(Transmission Control Protocols).
- ▶ Connection-oriented services may be done following Circuit-switched connection:

A circuit-switched connection is a dedicated communication path established between two endpoints for the duration of a communication session, ensuring a constant, guaranteed bandwidth and connection quality.

Connection Less Service

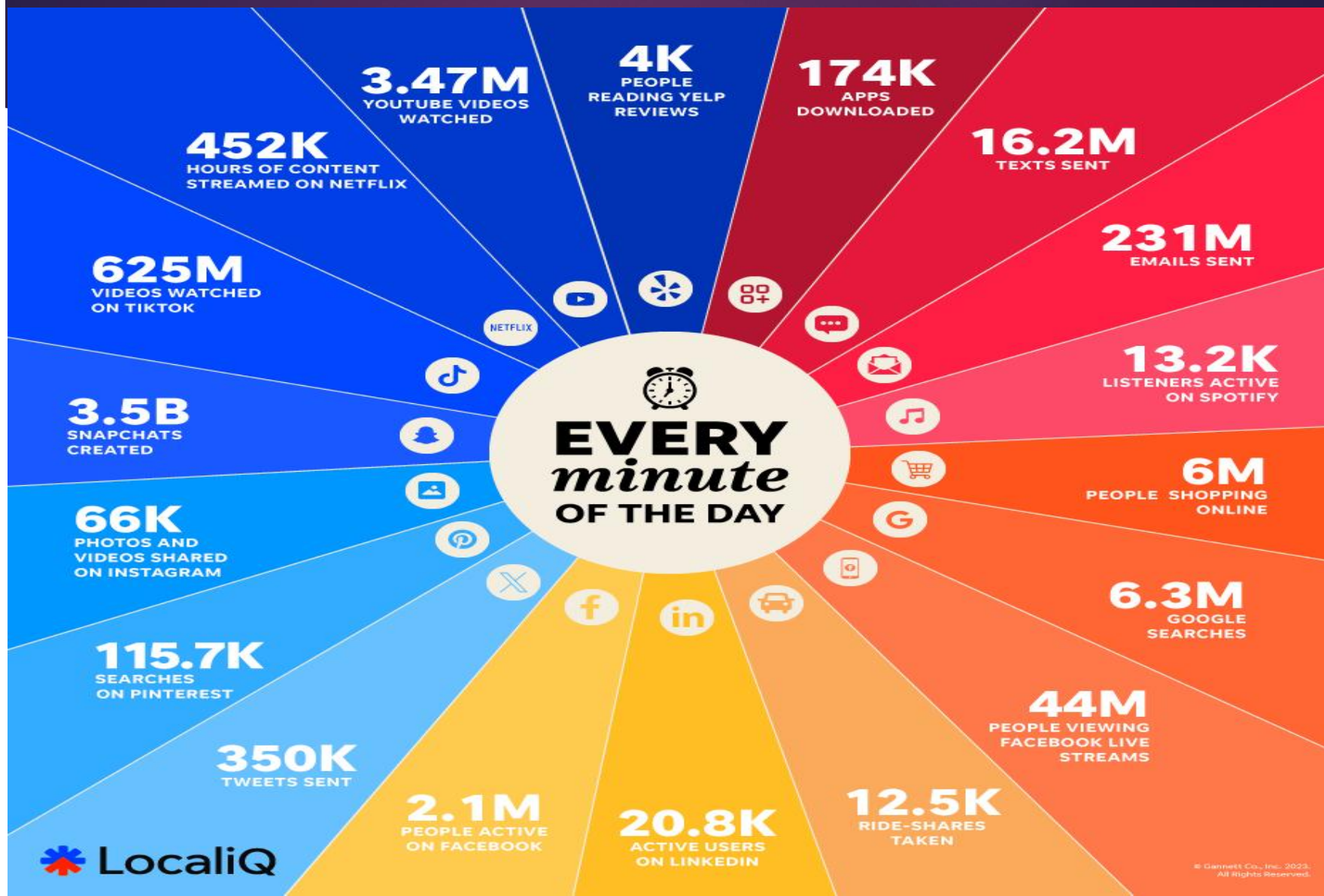
- ▶ It is similar to postal services, as it carries full address where the message is to be carried. Each message is routed independently from source and destination. The order of message sent can be different from order received.
- ▶ In connectionless the data is transferred in one direction from source to destination without checking that destination is still there or not or it is prepared to accept message.
- ▶ Authentication is not needed in Connection less service.
- ▶ In a connectionless service, the packets are sent from one party to another with no need for connection establishment or connection release.
- ▶ The packets are not numbered; they may be delayed or lost or may arrive out of sequence. There is no acknowledgment either.
- ▶ Common examples of Connection less services is UDP services.

Parameter	Connection-Oriented	Connection-Less
Connection	Prior connection need to be established.	No prior connection is established.
Resource Allocation	Resource need to be allocated.	No prior allocation of resource is required.
Reliability	It ensures reliable transfer of data.	Less reliable.
Congestion	Congestion is not at all possible.	Congestion can occur likely.
Transfer mode	It can be implemented either using Circuit switching or VCs.	It can be implemented using Packet Switching.
Retransmission	It is possible to retransmit the lost data bits.	Not possible
Suitability	It is suitable for long and steady communication.	It is suitable for bursty transmissions.
Signalling	Connection is established through process of signalling.	There is no concept of signalling.(Three step handshaking)
Packet travel	Sequential fashions	Random manner
Delay	More delay(Due to acknowledgement process)	NO delay due to absence of connection established phase.

1.9 Internet ,ISPs,Backbone Network Overview

Internet

- ▶ The internet is a **global network of interconnected computers and computer networks**. Internet offers:
 - ▶ **Global Connectivity:** The internet connects millions of devices worldwide, **facilitating communication and data exchange** on a global scale.
 - ▶ **Decentralized Structure:** It operates on a decentralized architecture, meaning **there is no central control or single point of failure**. Data is spread out over numerous servers and networks.
 - ▶ **Protocols and Standards:** The Internet **relies on standardized protocols** (e.g., TCP/IP) and open standards **to ensure compatibility and seamless communication between diverse hardware and software**.
 - ▶ **Diverse Services:** The Internet provides a diverse range of services, including **web browsing, email, file sharing, social networking**, and more. Users connect through Internet Service Providers (ISPs) using various technologies like **broadband**(highspeed internet access), **wireless**, or **dial-up**(using a telephone line)



ISPs(Internet Service Providers)

- ▶ An Internet Service Provider (ISP) is a **company or organization that gives users access to the Internet.**
- ▶ ISPs can vary in type—commercial, community-owned, non-profit, or privately owned.
- ▶ ISPs offer services such as **internet access, domain name registration, web hosting, and internet transit.**(provides access to the wider internet, often connecting customers to other ISPs.)
- ▶ ISPs are categorized into Tier 1, Tier 2, and Tier 3 based on their network infrastructure and level of connectivity:
 - Tier 1: ISPs have global networks and do not pay others for internet access.
 - Tier 2: ISPs connect to Tier 1 networks and may peer with others.
 - Tier 3: ISPs rely on Tier 2 or Tier 1 providers to reach the internet.

Tier 1 ISPs – The Internet Backbone

- Top-level **global providers** that do not pay anyone for internet access.
- **Interconnect with each other through peering** agreements (dashed lines).
- **Provide internet transit** (solid lines) to Tier 2 ISPs.
- Examples: AT&T, Verizon, Tata Communications.
- Role: **They build and maintain the core infrastructure of the internet across continents.**

Tier 2 ISPs – Regional/National Providers

- Purchase transit from Tier 1 ISPs for global internet access.
- Peer with other Tier 2 ISPs to exchange traffic and reduce costs.
- Sell internet services to Tier 3 ISPs and large customers.
- Examples: Airtel, Comcast, Vodafone.
- Role: They act as a middle layer, connecting the backbone to local ISPs and businesses.

Tier 3 ISP – The Local Access Provider

- Buy internet from Tier 2 (and sometimes Tier 1) ISPs.
- Deliver last-mile access to homes, small businesses, and communities.
- Focus on customer support and local services.
- Examples: Classic Tech, Vianet.
- Role: They are the final link to the end users.

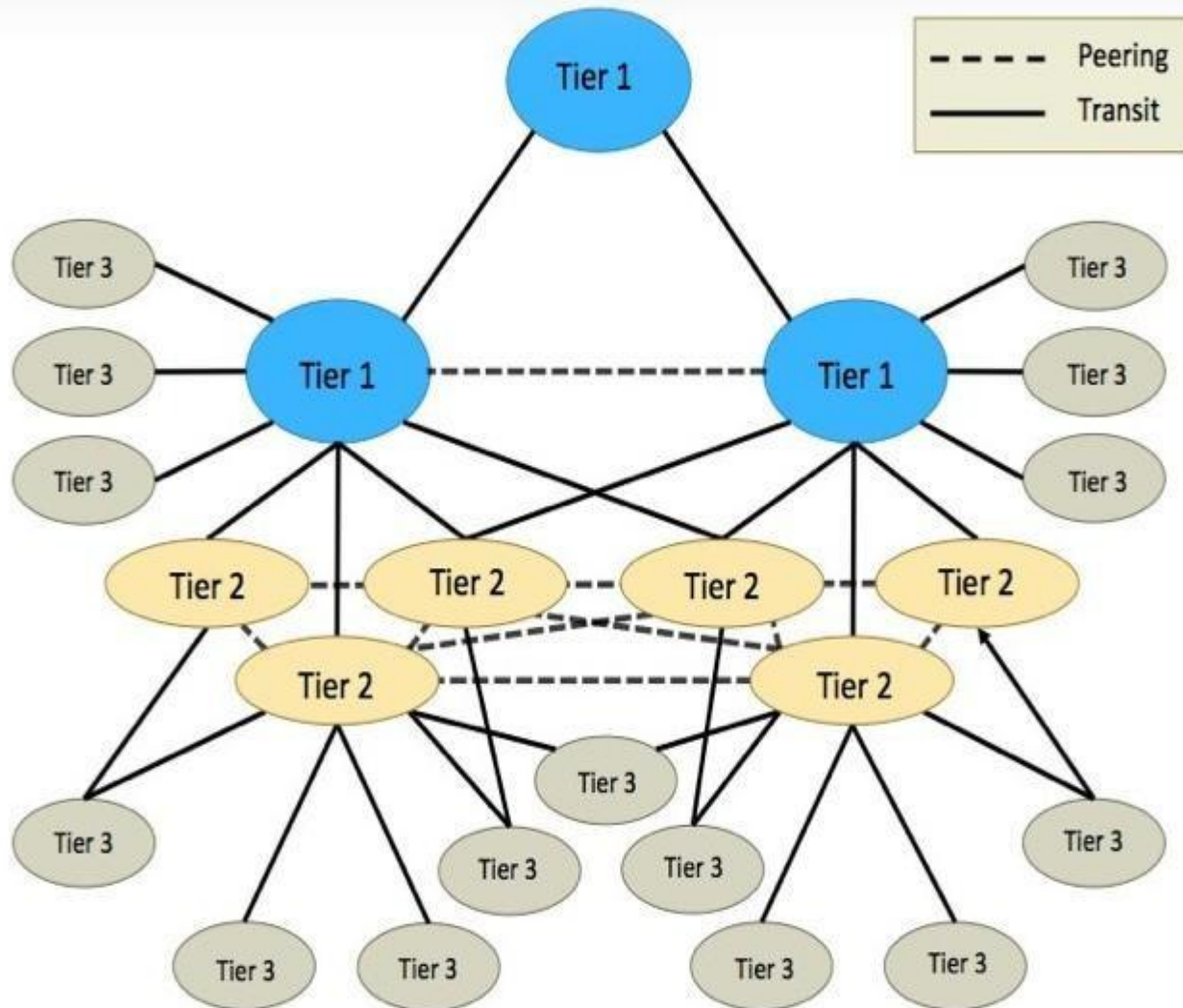
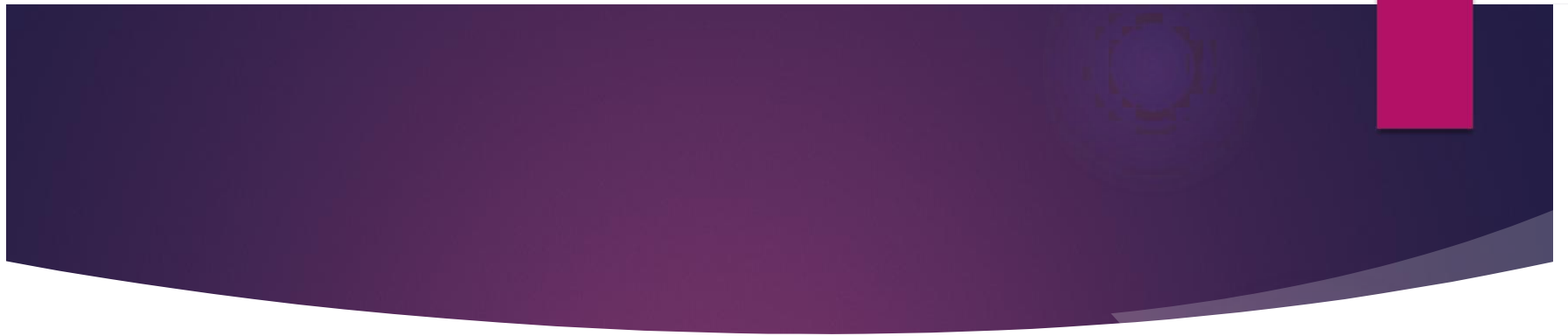


Fig: Tier Hierarchy of ISPs



Tier	Scope	Pays for Transit	Provides Transit	Peering	Examples
Tier 1	Global	No	Yes	Full	AT&T, Tata, NTT, Verizon
Tier 2	National/Regional	Yes	Yes (to Tier 3)	Some	Airtel, Vodafone, Cogent, Comcast
Tier 3	Local/Community	Yes	No	No	Classic Tech, Vianet

Types of ISP

► Hosting ISPs:

- Hosting Internet Service Providers specialize in **providing web hosting services, offering server space and infrastructure** for individuals and businesses to host websites and applications on the internet.

Example: GoDaddy – Offers web hosting, domain names, and email services.

► Mobile box ISPs:

- Mobile Box Internet Service Providers typically refer to telecommunications companies that provide internet services **through mobile networks(3G, 4G, 5G)**. Users connect to the internet using mobile devices, such as smartphones and tablets.

Example: Ncell or NTC (Nepal) – Offer mobile data packs and Wi-Fi routers (mobile boxes).

► Transit ISPs.

- Transit Internet Service Providers operate at the core of the internet, **providing high-level connectivity between different networks**. They **facilitate the transit of data between networks, often through extensive peering agreements with other ISPs**.

7/9/2025 Example : Tata Communications – Operates global undersea fiber and IP transit services. ©NTC

Cont.

▶ Virtual ISPs.

- ▶ Virtual Internet Service Providers are **companies that lease network infrastructure from other ISPs rather than owning their own physical network**. They offer internet services to end-users without owning the underlying network infrastructure.

▶ Free ISPs.

- ▶ Free Internet Service Providers offer internet access to users at **no direct cost**. These services are often **ad-supported, generating revenue through advertising** to offset the expenses of providing free internet access.

Ncell – Free Wikipedia Access (Nepal): Ncell used to offer free browsing of Wikipedia, especially for students and educational purposes.

▶ Wireless ISPs.

- ▶ Wireless Internet Service Providers deliver internet connectivity to **users using wireless technologies, such as Wi-Fi or fixed wireless access**. These services are often deployed in areas where traditional wired connections may be challenging or unavailable.

WorldLink (Nepal) – Provides wireless internet via Wi-Fi in many places.

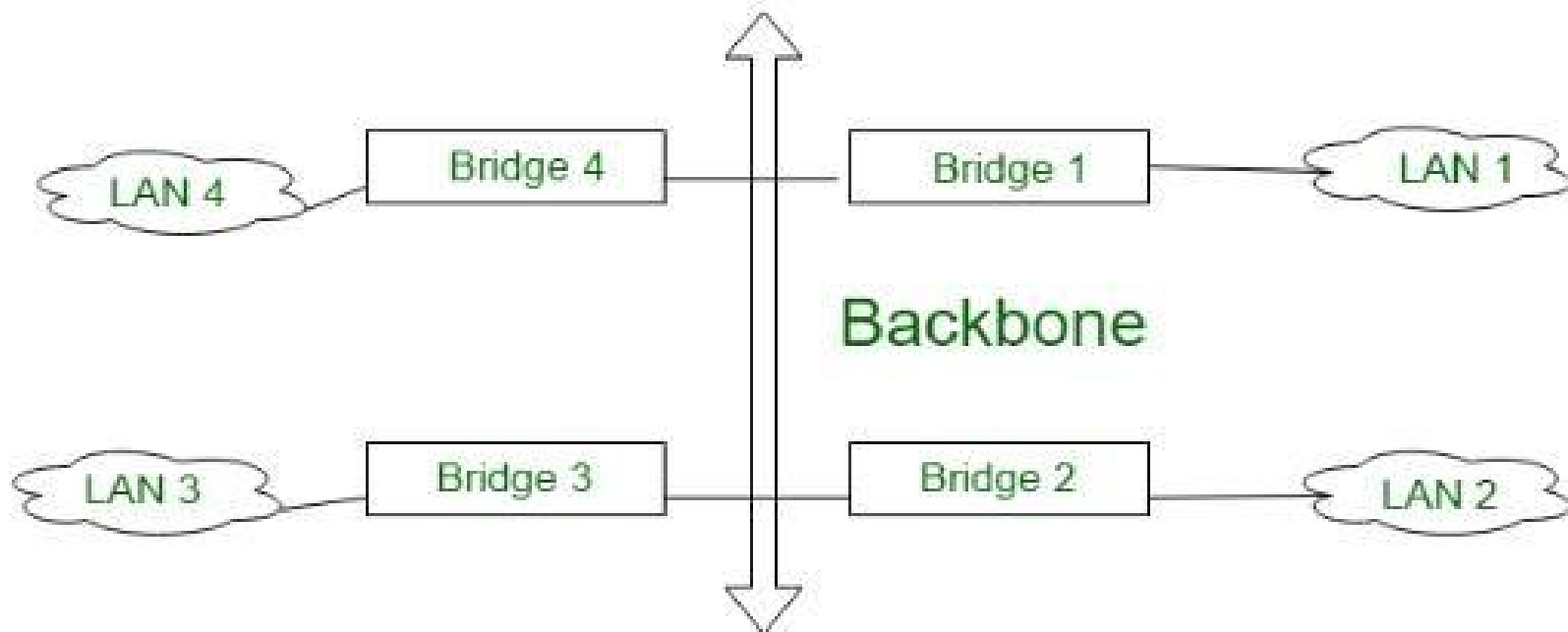
Backbone Network

- ▶ A Backbone Network, often referred to as a core network, is a high-capacity infrastructure that serves as the primary pathway for data transmission between different network segments and major locations within a network.
- ▶ It forms the central and essential part of a telecommunications or computer network, providing a robust and efficient means of interconnecting various subnetworks.
- ▶ The Backbone LAN serves as a high-capacity interconnection between the individual LANs in different buildings or departments. It is designed to handle the aggregated data traffic from multiple sources and facilitate efficient communication between the various LANs.

Types of Backbone

- ▶ The backbone **itself is a LAN**, it uses a LAN protocol such as Ethernet, Hence each connection in the backbone is itself another LAN.
- ▶ The commonly used architectures are:
 - ▶ Bus backbone
 - ▶ Star backbone
 - ▶ Interconnecting a remote LAN
- ▶ Bus backbone
 - ▶ In Bus backbone the topology used for the backbone is bus topology.
 - ▶ Bus backbones are normally **used as a distribution backbone to connect different buildings in an organization.**

Cont.

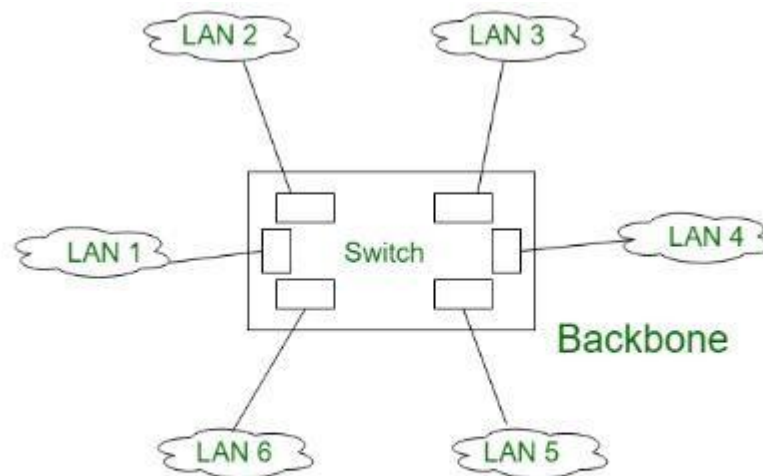


Structure of a Bus
backbone

Star Backbone

- ▶ The topology of this backbone is star topology.
- ▶ In a star backbone, sometimes called **switched backbone**, the topology of the backbone is a star. In this configuration, the backbone is just one switch that connects the LANs.
- ▶ In this configuration, the **switch does the job of the backbone and at the same time connects the LANs.**

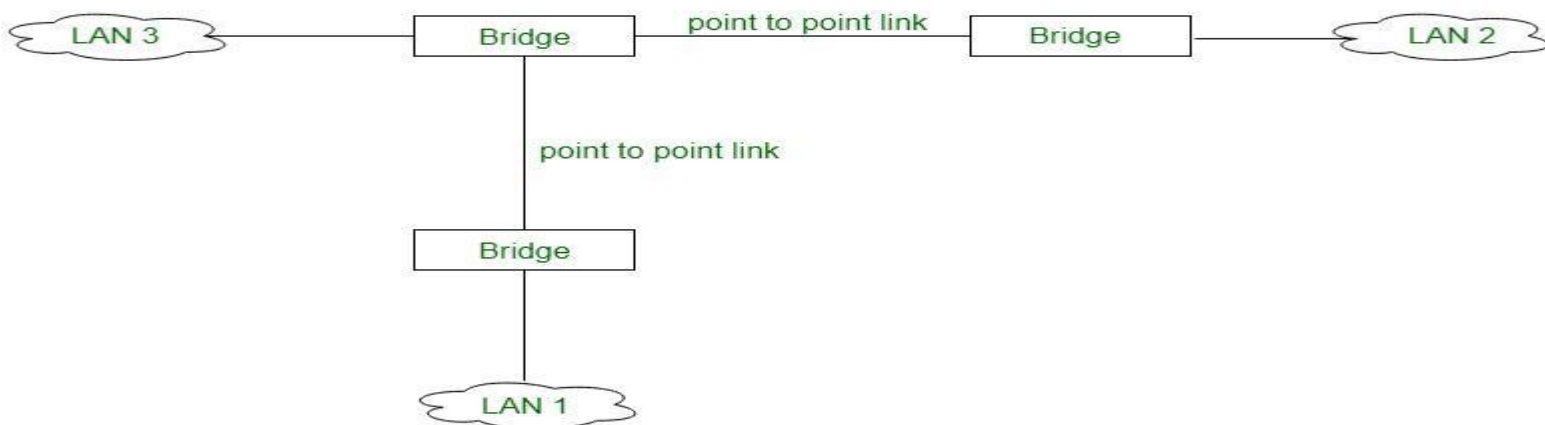
Cont.



Star Backbone

Interconnecting a Remote LAN

- In this type of backbone network, connections are established through remote bridges, serving as connecting devices that link LANs via point-to-point network links.



Connecting remote LANs to each other

Assignment 1

1. Explain different types of network topologies.
2. Explain different types of network types.
3. Define Protocol. Why do we need standards?
4. Explain internet tire or level starting from your home.
5. What is TCP/IP reference model and how is it different from OSI reference model?

Thank you!

Story 1: “The Coffee Handshake”

Arjun had noticed Meera over several weeks — the way she smiled at the barista, the quiet confidence she carried while discussing projects, and how her laughter lit up the room during group study. But they never really talked.

One evening, after rehearsing his words in his head a dozen times, Arjun finally took out his phone and typed a message:

“Hey Meera :) I hope this isn’t random... but would you like to grab coffee sometime?”

His thumb hovered over the send button, then he tapped it, heart pounding.

He didn’t just send the message and forget — he waited, eyes glued to his screen.

Minutes later, his phone buzzed.

“Hi Arjun :) Not random at all. I’d love to! Are you free this weekend?”

Relief and excitement washed over him.

He replied quickly:

“Saturday works great. There’s a quiet café near the park — I think you’ll like it.”

Her reply came almost instantly:

“Sounds perfect :) Looking forward to it!”

Through this back-and-forth, a connection was formed — not just of messages, but of trust, anticipation, and mutual understanding.

Each message was acknowledged, each response carefully crafted, and the conversation flowed in an orderly sequence.

There were moments of pause and delay, but they were part of the process — ensuring the conversation stayed meaningful and reliable.

This steady exchange laid the foundation for a real connection — something lasting.

Story 2: “A Note in the Wind”

Ayan had admired Rhea from afar for months. She was always reading by the window in the library, lost in her own world.

He never approached her. He didn't want to disturb the quiet moment or risk awkwardness.

One day, on a whim, he scribbled a small note:

“Hi Rhea, I don't know if you even know me. But I just wanted to say you have a peaceful kind of beauty. No reply needed.”

He folded the note carefully and slipped it into a friend's notebook, trusting the friend to pass it on.

After that, he didn't ask if she read it.

He didn't check for a reply.

Maybe she saw it, maybe not. Maybe she tossed it away or never got it.

For Ayan, it didn't matter. The message was sent.

No waiting, no confirming — just a one-way gesture.

Thank you!

